

Marcus Ruth

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EDUCATION

University of Central Florida
Bachelor of Science in Computer Science

Aug. 2024 – May 2027
Orlando, FL

Valencia College
Associates of Arts - General Studies

Aug. 2022 – May 2024
Orlando, FL

RELEVANT EXPERIENCE

University of Central Florida
Teaching Assistant - Data Structures & Algorithms (COP 3502C)

Jan 2025 – Present
Orlando, FL

- Lead lab sessions and office hours covering data structures (trees, heaps, hash tables), instructing over 100 students per semester.
- Designed programming assignments targeting specific learning goals such as recursive thinking, dynamic memory management, and reasoning about augmented data structures.
- Grade assignments and provide constructive feedback to help students improve algorithmic problem-solving and debugging of C-based implementations.

Downtown Diner Celebration
Food Server

Jan 2024 – Present
Celebration, FL

- Directing tourists to desired areas to travel to in local vicinity.
- Exceptional multitasking skills due to larger responsibility in restaurant.
- Efficiently managing time between customer requests and job responsibilities.

Mindhome
Software Engineer - Internship

December 2022 - February 2023
Remote

- Completed ROS2 training utilizing Python, building foundational skills in AI and robotics programming.
- Contributed to backend development in Ruby, implementing API endpoints and assisting with database integration to support core platform functionality.
- Participated in code reviews and collaborated with senior engineers to refine application logic and improve system reliability.
- Developed a holistic understanding of software development methodologies including Agile workflows and version control practices.

PROJECTS

Nudge | *Next.js, TypeScript, tRPC, Zod, Supabase, MediaPipe, LiveKit, Tailwind CSS*

- Developing a student accountability platform that pairs college students for live video-call study sessions with real-time distraction tracking.
- Integrated MediaPipe computer vision on the client side to detect off-screen gaze and phone usage during study sessions.
- Designed backend architecture with Supabase for authentication (.edu email gating), Postgres database, and realtime session synchronization.
- Implemented solo and duo session modes with structured timer workflows and AI-generated end-of-session productivity reports.

miniredis | *C++, POSIX sockets, systems programming*

- Built a Redis-like in-memory key-value store from scratch using modern C++ and POSIX sockets.
- Implemented event-driven TCP server with poll() capable of handling multiple concurrent client connections.
- Developed RESP protocol serialization and core commands (SET, GET, PING, ECHO) with per-client buffering.

TECHNICAL SKILLS

Languages: Python, Java, C/C++, Go, TypeScript, JavaScript, SQL

Web: Next.js, React, tRPC, Zod, Supabase, Postgres

Libraries: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, Nixtla

Developer Tools: Git, VS Code, Unix/Linux, AWS, PostgreSQL, Neovim